RIKESH MANDAL

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PROFESSIONAL PROFILE

MSc Artificial Intelligence graduate (Distinction, 82%) with a First-Class Honours degree in Computer Networking and IT Security, and professional experience in software development within the FinTech industry. Skilled in Python, Java, Spring Boot, REST APIs, SQL, Docker, and PyTorch, with hands-on experience developing backend applications, machine learning models, and full-stack web solutions. Brings an interdisciplinary background in software engineering, AI, and 3D computer graphics, with strong problem-solving abilities and experience translating technical requirements into practical solutions. Passionate about building reliable, scalable software and applying emerging technologies to real-world challenges.

KEY SKILLS

Programming Languages: Python, Java, JavaScript, SQL, MATLAB

Backend Development: Spring Boot, Flask, Django, REST API Development, SQLAlchemy, Jinja2

Databases, Search & Caching: MySQL, PostgreSQL, SQLite, MongoDB, Redis, Elasticsearch

APIs & Messaging: OpenAPI/Swagger, NATS, WebSockets, JSON, Event-Driven Architecture

DevOps & Deployment: Docker, Docker Compose, Linux (Ubuntu), Nginx, Gunicorn, Git, GitHub

Testing & QA: Unit Testing, Integration Testing, Pytest, Postman, Debugging

Frontend Development: Vue.js, HTML, CSS

Machine Learning & Computer Vision: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Open3D

PROFESSIONAL EXPERIENCE

Assistant Java Developer, FocusOne Payment Solutions (Fintech) **2022 – 2024**

Backend systems powering real-time credit and debit card authorisation for a regulated fintech platform

- Designed and maintained high-performance REST APIs using Java and Spring Boot for a platform processing real-time card authorisations across thousands of daily transactions.
- Refactored legacy applications through root-cause debugging and targeted code improvements, reducing production incidents and improving system reliability.
- Evaluated NATS messaging for event-driven architecture adoption and presented findings that shaped the team's infrastructure roadmap for low-latency, fault-tolerant systems.
- Built and maintained Swagger/OpenAPI documentation and contributed to API endpoint testing using Postman, cutting onboarding time and improving cross-team collaboration on API contracts.
- Delivered new backend features aligned with evolving product requirements in a fast-paced Agile environment, participating in code reviews and sprint planning.
- Authored system specifications and API design documentation supporting long-term maintainability, reliability, and regulatory compliance.

KEY PROJECTS

Microblog — Full-Stack Flask Web Application | GitHub | Live Demo | API Documentation

Developed and deployed a Flask web application with REST APIs, MySQL, full-text search, multilingual support and Redis-backed rate limiting on an Ubuntu home server.

- Developed REST APIs for user accounts, posts, messages and followers, handling account management, secure token-based access and protected routes. Included consistent JSON responses, pagination and standard HTTP status codes to make the API easy to integrate with and maintain.
- Designed and maintained the MySQL database using SQLAlchemy, defining relationships between users and posts, managing database migrations, and improving data retrieval through indexing and paginated queries.
- Deployed the application on an Ubuntu home server with Gunicorn, Docker Compose, Nginx and Tailscale Funnel, providing public HTTPS access. Kept MySQL data persistent and supporting services in separate containers.
- Integrated Elasticsearch to add full-text search, using secure HTTPS connections and providing a way to rebuild search indexes whenever the search data needs to be refreshed.
- Added multilingual features using LibreTranslate and Flask-Babel, enabling post translation and interface language support while running translation as a separate containerised service.
- Produced OpenAPI 3.0 documentation covering authentication, request and response formats, pagination and errors; published an interactive reference with Scalar and GitHub Pages.
- Implemented Redis-backed rate limiting for login, registration and API routes to control repeated requests and return HTTP 429 responses. Configured Nginx to pass the correct client IP.

Distributed Messaging System with NATS.io (Java Microservices)

- Led the design and implementation of a distributed microservices communication system using NATS.io, taking responsibility for architecture, message design, and system integration strategy.
- Built pub-sub based request-reply workflows in Java, handling concurrent request processing via multithreaded responders to improve responsiveness under load.
- Implemented queue groups to balance requests across multiple service instances, improving reliability and preventing bottlenecks.
- Defined JSON-based message contracts enabling independently built services to communicate without tight coupling and tested the system under concurrent load to validate stability.

MSc Artificial Intelligence Dissertation - "An AI-Driven Framework for Automated 3D Model Generation from 2D Images" GitHub | Dissertation | Live Demo

Deep Learning | Computer Vision | 3D Reconstruction

As part of my MSc Artificial Intelligence dissertation, I independently designed and implemented an end-to-end pipeline for reconstructing 3D models from a single RGB image. The project combined deep learning-based depth estimation with classical geometric processing techniques, covering the complete workflow from data preprocessing to 3D mesh generation.

- Designed and implemented an end-to-end image-to-3D reconstruction pipeline using Python and PyTorch.
- Fine-tuned a ResNet-50 encoder-decoder architecture for monocular depth estimation using the NYU Depth V2 dataset.
- Developed custom data preprocessing and loading pipelines, including image resizing, normalisation, batching, and dataset management for efficient model training.
- Trained and evaluated the depth estimation model, monitoring performance through quantitative metrics and qualitative visualisation of predicted depth maps.

- Generated dense point clouds from predicted depth maps by combining depth information with the corresponding RGB images.
- Reconstructed 3D meshes from generated point clouds using Poisson Surface Reconstruction, followed by normal estimation and mesh refinement to improve reconstruction quality.
- Built a modular pipeline separating data preprocessing, model training, inference, point cloud generation, and mesh reconstruction, improving maintainability and reproducibility.
- Researched the integration of deep learning and geometric processing techniques for generating meaningful 3D representations from monocular images.

Sign Language-to-Speech Glove **GitHub**

- Built a hardware-software IoT system using an Arduino glove with flex sensors and a gyroscope to capture hand gestures and trained a KNN classifier on a self-collected dataset to translate ASL gestures into real-time speech.
- Tech stack: Python, Arduino, Scikit-learn, IoT sensors.

Animal Image Classification System **GitHub**

- Designed a custom CNN from scratch (no pre-trained weights) to classify 9 animal classes, achieving 79.51% accuracy using progressive convolutional depth, 3x3 filters, batch normalisation, and dropout.
- Tech stack: Python, PyTorch, OpenCV, NumPy, Scikit-learn.

EDUCATION

MSc Artificial Intelligence — Distinction (82%) *University of East London* **2024 – 2025**

BSc Computer Networking & IT Security — First Class (81.94%) *London Metropolitan University* **2021 – 2023**

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Data Analytics Specialisation — Coursera (2023)
- API Designer & API Security Architect — API Academy
- AWS Academy Cloud Foundations — Amazon Web Services
- Deep Learning Onramp & Machine Learning Onramp — MathWorks
- Data Science Foundations — LinkedIn Learning